PROJECT 01 / EV3 NAVIGATION

Build strategy and parts

A repeatable driving base, with an explicit assembly and measurement sequence.

How to use this guide

Build the official Driving Base first, then the gyro module. Add color sensing only after baseline tests pass. Those manuals supply exact LEGO parts, holes, pin positions, and axle lengths. This companion supplies the build order, integration checks, original layout sketches, and calibration procedure.

The sketches are functional schematics, not LEGO CAD or dimensioned fabrication drawings. The custom combined layout is a proposed arrangement: check available holes, parts, and clearance physically. No extra-kit parts are assumed available. Keep the official geometry unless a measured problem justifies changing it.

Gather before assembly

Core: 1 EV3 brick with battery; 2 large motors; 2 matching wheels and tires; the kit ball caster assembly; the beams, axles, bushes, connectors and pins specified in the base manual.

Sensing: 1 EV3 gyro; 1 EV3 color sensor; the mounting parts specified in both module manuals; 4 cables total for 2 motors and 2 sensors. Choose lengths that reach without dragging.

Integration: reserve leftover beams and friction pins for bracing and cable retention. Check the module parts callouts against what remains after building the base; the combined parts budget and simultaneous attachment points have not been physically verified.

Tools for testing: ruler or tape measure, square, representative floor surface, temporary test marks, notebook, and a laptop for logging. Do not add marks to the actual course without permission. A second color sensor is an optional later upgrade, not required.

Assembly references: use these alongside the companion

All three manuals below are official LEGO documents. Follow their illustrated sequences in full; they are linked, rather than reproduced in this repository.

1. Driving Base / 46 PDF pages [open official source]

2. Color Sensor Down / 7 PDF pages [open official source]

3. Gyro Sensor / 10 PDF pages [open official source]

Minimum build and decision order

Review revision / adapt the official base before considering a new chassis.

Build A: the engineering baseline

Use the unmodified official Driving Base, two large motors and encoders, battery, ball caster, and official gyro mount. Secure its three cables. Do not add a second color sensor, ultrasonic scanner, extra gear train, or custom chassis initially.

Do not move the brick or change wheel supports to imitate the layout sketches. Reseat loose pins first. Add bracing only where a specific joint visibly moves; otherwise it can introduce twist. The assignment does not require a goal sensor.

Gate 1: prove the base before accessories

Confirm motor/gyro signs, repeatable starting alignment, stable gyro startup, free caster motion, and no wheel rub. Repeat straight drives, both turn directions, and complete routes at realistic durations. Record independent physical errors. Ten trials are an initial screen, not a reliability certificate.

Compare smooth low and moderate speeds. Extremely slow motion extends run time and gyro-bias exposure; choose the better measured full-route performance. Slow for turns and endpoints. Keep pauses brief and purposeful.

Build B: optional color sensing

Add the official downward module only if it fits alongside the gyro with available parts and preserves clearance. Its first job is goal confirmation in the mapped goal region. It cannot protect the full robot width from flat obstacles.

Start with seam correction DISABLED. Evaluate it only if route error exceeds clearance and actual seams are detectable. A failed seam experiment must not delay baseline testing. Recalibrate and rerun routes after payload/mount changes.

The combined mounts and parts budget remain unverified. If modules conflict, retain Build A. Do not improvise a fragile bracket just to complete Build B.

When a task-specific mechanical modification is justified

If the sensor boom dominates turning radius, shorten its mount while retaining lens clearance and a rigid attachment, then remeasure offsets and R. If a joint flexes, brace that joint. If the caster snags seams, inspect its clearance and loading. Change one thing at a time and repeat the failed test.

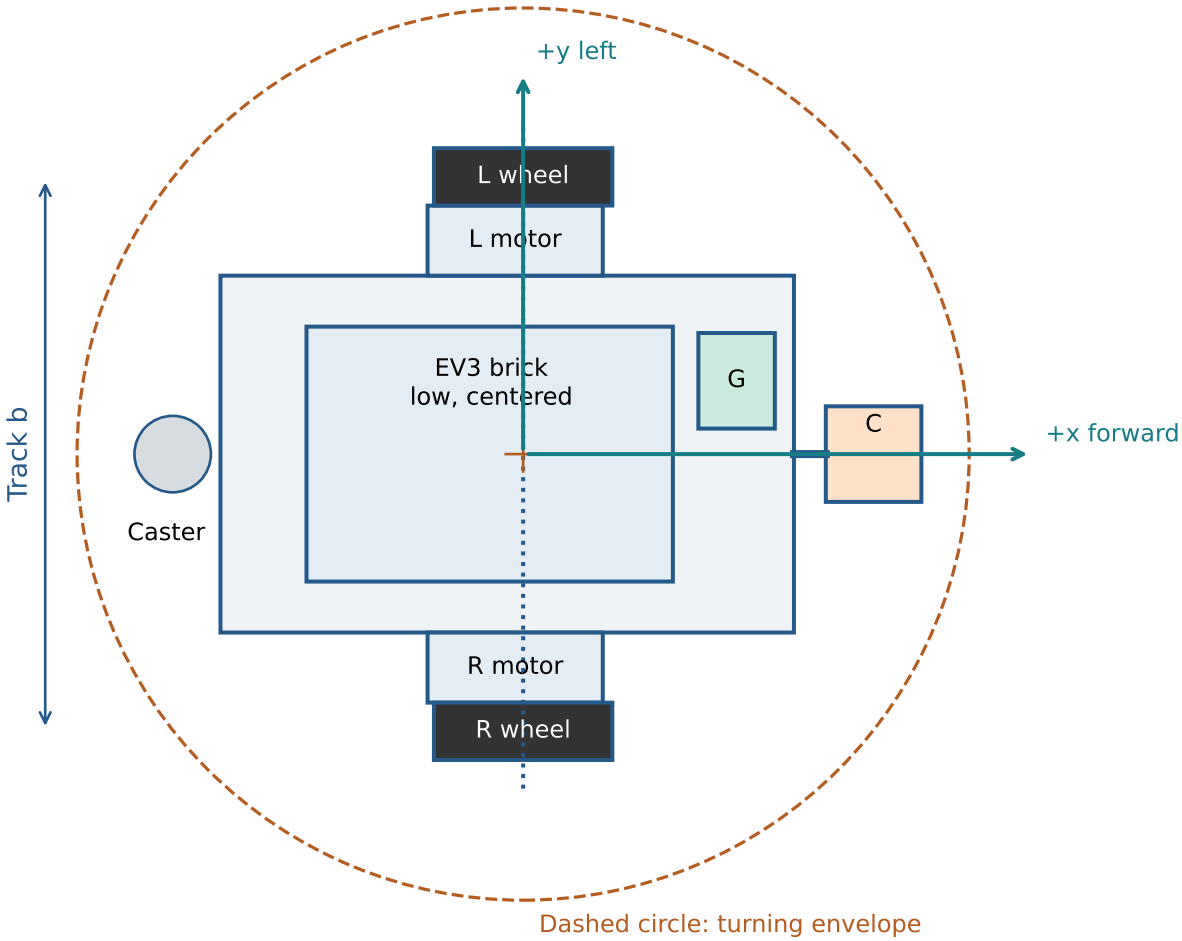
A new chassis is justified only when the stock geometry cannot meet measured clearance or traction needs. Until then, stock geometry is the simpler baseline. Custom hole-by-hole instructions require a verified parts inventory and physical fit.

Limits of the baseline

Gyro plus encoders provide finite-run dead reckoning. They cannot recover all wheel slip or guarantee every arbitrary course. The assignment acknowledges occasional failures; our acceptance depends on repeated full-route tests and clearance reserve, not on sensor specifications alone.

01 / Top-view layout

Functional layout / not to scale / use official manuals for exact LEGO connections.



Reference points

O is the midpoint between the drive-wheel contact centers. Define robot pose here; do not silently use the brick center or front sensor as the position reference. +x is forward; +y is left; positive heading is counterclockwise when viewed from above.

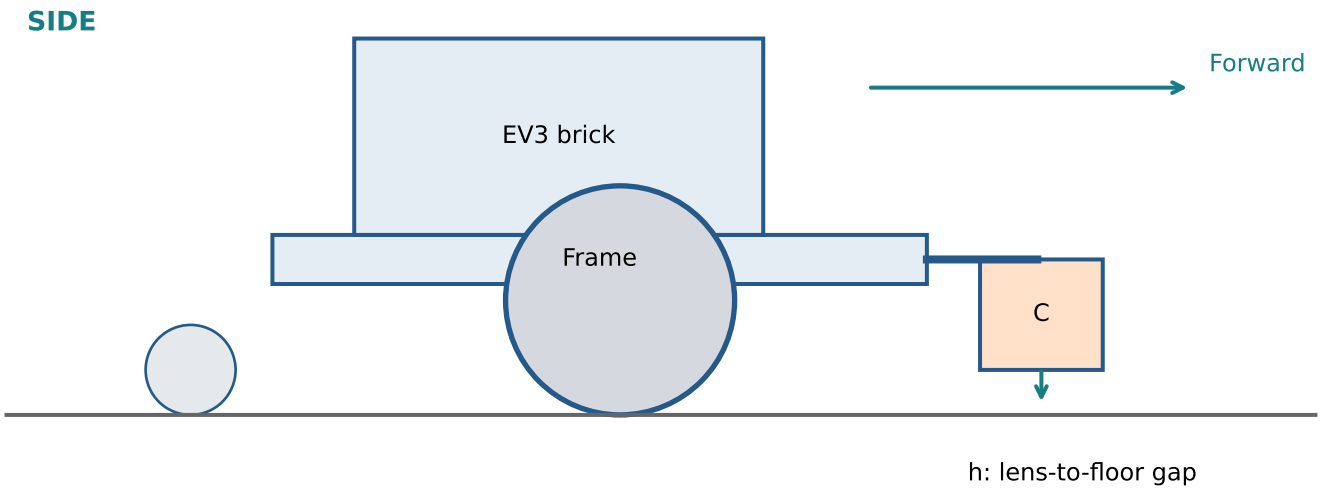
G = gyro on a rigid mount; exact position may differ. C = OPTIONAL downward color sensor, drawn on the centerline for clarity. If the official module is offset sideways, keep it and measure that offset. Do not rebuild a sound module just to match this sketch.

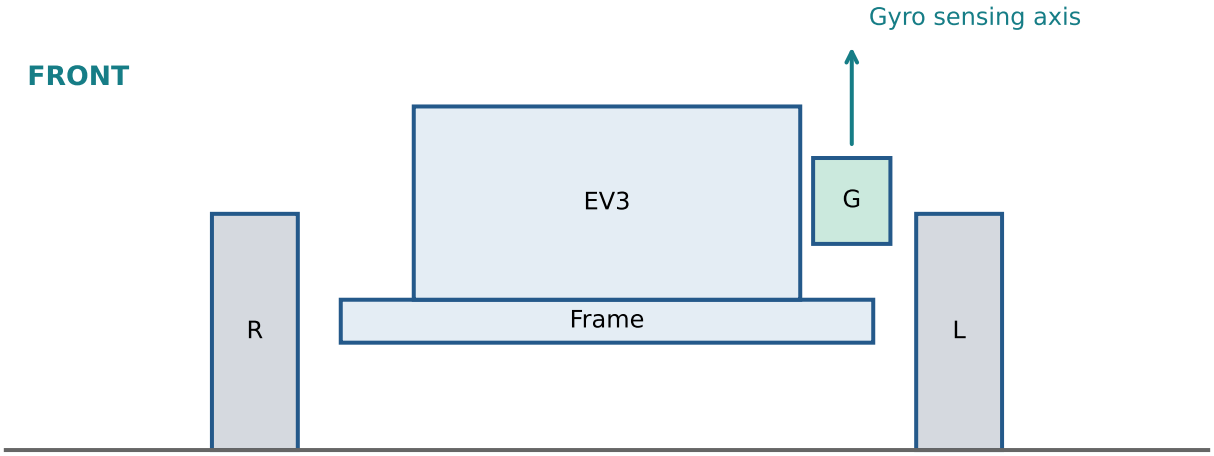
Clearance rule

Measure R from O to the farthest protruding point, including cables. Use a circle of radius R plus an error allowance for conservative planning and in-place turns. A 0.61 m corridor admits this envelope only when 2(R + allowance) < 0.61 m. This is a design check, not a measured claim about the base.

02 / Side and front views

Sensor orientation and support geometry / conceptual positions / no fixed dimensions.





Mounting checks

Both driven tires must rest on the floor without chassis twist. The caster must roll freely and cross tile seams without lifting a driven wheel. Leave clearance so wheels do not rub axles, beams, cables, or sensor brackets.

Orient the gyro to measure yaw (turning about vertical); follow the official module. The case position in this drawing does not define its internal sensing-axis orientation. A hand-rotation test must confirm it.

Mount the color lens perpendicular and close to the floor without touching it. Try a 5-10 mm lens gap as an experimental starting range, then adjust using real readings and seam clearance. This range is a tuning suggestion, not a LEGO specification.

03 / Assemble the driving base

Complete each checkpoint before adding sensors or changing geometry.

A. Inventory and prepare

1. Open the Driving Base manual from page 1 of this guide. Gather each part from its illustrated callouts. Match axle lengths using the manual guides; count beam holes.

2. Inspect the two tires. Use a matching pair with the tires fully seated. Clean off dust and inspect for damage. Prepare a charged battery and keep that battery configuration through calibration and demonstration.

B. Build the stock base

3. Follow the complete official illustrated sequence. Keep the specified motor connections, wheel supports, brick orientation, and caster geometry. Compare left and right sides after each assembly stage. Do not substitute axle lengths casually.

4. Before fitting the wheels, check that each axle and retaining bush is seated as shown. After fitting them, check lateral play and rubbing. Do not overtighten a bush against a rotating part.

5. Install the battery and place the base on the target floor. Press gently on each corner. Look for a lifted tire, loose joint, or twisted frame; correct the assembly before proceeding. Rocking on an uneven seam needs a separate floor test.

C. Reinforce only if necessary

6. Hold the motor bodies and gently try to twist the frame. If the motor spacing changes visibly, identify the moving joint. First reseat its existing pins.

7. If spare parts permit, add a cross-member between existing left/right frame mounting points, without shifting the stock geometry. Use two separated friction-pin connections at each end where holes permit; a single pin can act as a hinge. This is a fit-to-build modification, not a verified hole-specific instruction.

8. Retest the wheels and caster after bracing. Reject any brace that bends the base, rubs a moving part, obstructs the battery, or extends the turning envelope unnecessarily.

Base acceptance checkpoint

[ ] Wheels match and are seated. [ ] No tire or axle rubbing.

[ ] Motor/frame joints do not shift visibly under light hand load.

[ ] Both driven tires contact the floor. [ ] Caster moves freely.

[ ] Brick buttons, battery access, and cable sockets remain accessible.

04 / Add and validate sensors

Integrate one module at a time; preserve a working mechanical baseline.

D. Gyro assembly

9. Build the official Gyro Sensor module using its illustrated sequence. Connect it to the driving base at the specified attachment points. The mount should be short, rigid, and secured against pivoting. Do not add a flexible mast.

10. With motors stopped, keep the robot completely still while connecting the gyro and during startup. Read the angle and rotation rate if available. Log a stationary sample before moving. Simply setting the displayed angle to zero does not remove bias.

11. Turn the whole base counterclockwise by approximately 90 degrees using a square. Check that the gyro measures that turn, then turn back. If the sign is opposite your robot convention, invert it once in the hardware adapter. If the rotation is barely detected, check the mount orientation and sensor mode.

E. Downward color sensor

12. Build the official Color Sensor Down module. Before attaching it alongside the gyro, check for occupied holes, shared parts, cable interference, and wheel clearance. If the mounts conflict, test each official module separately first; the combined mount remains a physical integration task. Avoid improvised long cantilevers.

13. Position the lens downwards, perpendicular to the surface. Measure the floor gap with the full robot loaded. Slowly move it over a tile seam by hand with motors off. Nothing should scrape and the bracket should not bounce.

14. Record reflected-light values over plain floor, grout/seams, obstacle cardboard, and goal material, in actual room lighting. Test at more than one position. Use a fixed measurement mode during a run. Adjust the height only if separation or clearance is poor.

F. Secure cables and verify the combined assembly

15. Route cables along beams inside the footprint. Retain them with suitable LEGO openings or clips; do not pinch them with pins. Leave slack at each connector.

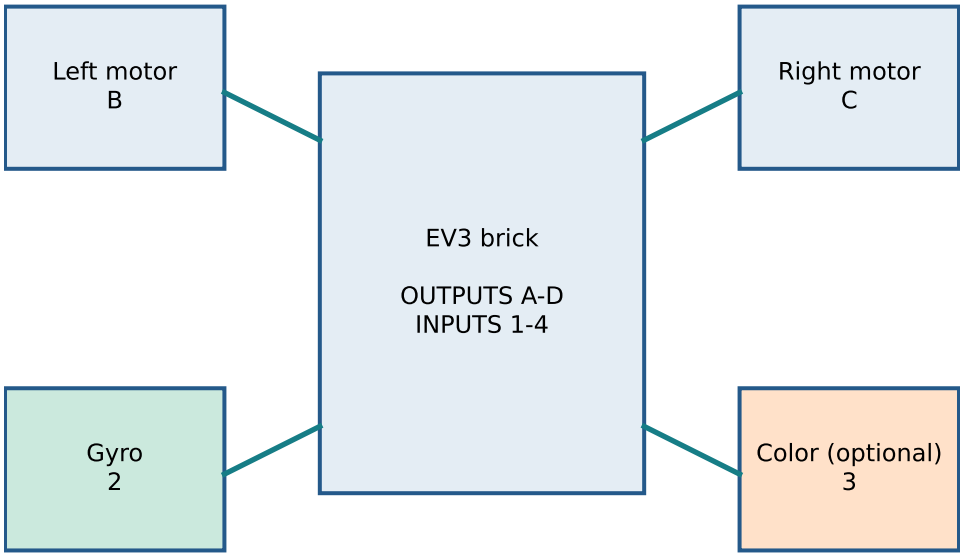
16. Roll and turn the unpowered robot by hand, then perform slow powered turns. Check for dragging cables, tire contact, caster snags, sensor vibration, and loose pins. Measure sensor offsets after the final mount is fixed.

Official gyro startup guidance [open official source]

Official color sensor mounting and light guidance [open official source]

05 / Wiring and sign conventions

Recommended project ports / these are choices, not mandatory LEGO assignments.



Connection map

B = left large motor; C = right large motor; 2 = gyro; 3 = optional color sensor. Left and right mean as seen from behind the robot, looking forward. If your official assembly is already cabled differently, either move the cables or change the project configuration; the two must agree. Leave unused ports empty initially.

Bench test with wheels safely clear of the floor

1. Command only B at low speed briefly; verify it is the left wheel. Stop it.
2. Command only C; verify it is the right wheel. Stop it.
3. Define positive logical speed as forward ground travel on both wheels. Mirrored motors may need opposite raw signs. Make the encoder signs match that convention too.
4. Confirm counterclockwise physical rotation produces positive logical gyro heading.
5. Verify the stop function and a reachable brick-button abort before floor tests.

Programming boundary

Use the motor/sensor API permitted by the instructor. LEGO EV3 MicroPython uses Pybricks, so a blanket library ban needs clarification before choosing imports. This build does not require a prebuilt DriveBase, navigation library, or sensor-fusion package. Keep raw access and sign conversion in one hardware adapter.

06 / Geometry and calibration

Measure the fully assembled robot on the demonstration floor.

Measurements to record

D_L, D_R: effective wheel diameters in meters. b: lateral distance between tire contact centerlines. R: farthest horizontal distance from wheel midpoint O to any robot part. s_x, s_y: color-lens offset from O, forward and left respectively. h: lens-to-floor gap.

Use actual measurements rather than copying the normalized drawings. Keep a record of battery, floor, speed, tires, and payload so later tests are comparable.

Distance scale: calibrate forward and backward

On a clear straight lane, align the robot with a square. Command a known wheel rotation at low speed and measure actual forward travel at O. Repeat at least five times and check lateral error. For n wheel rotations, D_effective = travel / (pi * n). This relation assumes a straight, nonslipping run. Exclude visibly slipping trials from scale calibration, but keep them in the reliability record.

Use separate wheel scale factors only when repeated tests support the difference. A gyro heading controller can hide wheel mismatch, so inspect the encoder difference needed to keep straight. Do not repeatedly retune from a single run.

Odometry equations and independent checks

With logical encoder changes dL and dR in degrees:

left_distance = pi * D_L * dL / 360

right_distance = pi * D_R * dR / 360

ds = (left_distance + right_distance) / 2

encoder_heading_change = (right_distance - left_distance) / b [radians]

For short update intervals, use ds and the midpoint heading to update x and y. Gyro heading can guide steering; compare its change with encoder heading as a consistency check. Disagreement flags possible slip or drift, but does not identify the true pose.

Turning calibration can use b_effective = (right_distance - left_distance) / measured_heading_change, with heading in radians. Use a square or independent angle reference as well as the gyro; do not calibrate everything against one unverified sensor.

Sensor offset matters

For robot heading theta, the lens position is:

lens_x = x + s_x*cos(theta) - s_y*sin(theta)

lens_y = y + s_x*sin(theta) + s_y*cos(theta)

Use the lens position when matching a seam or goal observation to the map. A forward sensor reaching black does not mean the robot center has reached the goal center.

07 / Prove the error budget

Suggested engineering targets / these are not assignment grading requirements.

Test 1: stationary gyro and a usable drift budget

Record heading over the expected run duration while the robot remains still. Repeat startup and initialization to detect inconsistent bias. Example: 3 degrees heading error over a 1 m straight produces about 52 mm lateral error (1*sin(3 degrees)). Choose an acceptable heading error from available path clearance, not a universal threshold. A stationary test does not capture every motion-induced error.

Test 2: straight runs and turn repeatability

Run ten 1 m straight trials, independently realigning before each. Record longitudinal error, lateral error, heading error, elapsed time, and visible slip. Repeat backward if the planned route uses reverse motion. Start around 0.05-0.10 m/s and tune on the floor.

Run ten 90-degree turns in each direction. Record final physical angle, gyro angle, encoder readings, caster behavior, and any center displacement. Start around 15-30 degrees/s and slow near the target. Speeds are initial tuning suggestions.

Drive a square in both directions and measure return-to-start error. Then test a representative long route; one successful short test is not enough.

Test 3: decide whether seams are usable landmarks

Sample plain tiles and seams at several locations and lighting conditions. Cross seams slowly at expected approach angles. Record missed detections and false positives. Enable corrections only if readings and mapped seam locations are consistently compatible.

One seam observation constrains one coordinate; it does not recover full pose. Match only the expected seam within a plausible pose window, accounting for sensor offset. If two seams could explain the reading, reject the correction.

Black cardboard and the black goal can resemble seams in reflected light. Require map context and multiple readings. Do not use obstacle crossings as landmarks: crossing obstacle tiles violates the task.

Clearance acceptance

Let E be a conservative position-error allowance derived from repeated full-route trials, with additional reserve; a sample maximum is not a guarantee. Require planned center clearance >= R + E from obstacles and boundaries. Check every connecting segment and every turn, not just grid nodes. If no route survives this conservative model, improve the build/localization or use a more accurate footprint model; do not silently shrink clearance. Gyro plus encoders cannot detect all slip or bound position error forever.

08 / Start the code in stages

Keep each stage independently observable before combining navigation.

Stage A: raw hardware adapter

Implement read_wheel_angles(), read_heading(), optional read_reflection(), set_wheel_speeds(left, right), and stop(). Normalize signs here. Log timestamp, both encoders, gyro, reflection, requested wheel speeds, and current motion state.

Check the stop button in every control loop, stop on exceptions with try/finally, and include motion timeouts. Detect lack of encoder progress when movement is commanded. A slip/stall abort invalidates the pose estimate: stop and physically realign before restarting unless an independent localization method has recovered pose.

Stage B: movement controller

Implement turn_to_heading(target) and drive_distance(distance, target_heading). Run a timed feedback loop; begin with a simple proportional heading controller and conservative speed limits. Tune the loop period from measured I/O timing.

Use absolute map headings, not a sequence of relative 90-degree commands. For forward motion, positive heading error means a left turn: reduce the left wheel speed and increase the right wheel speed. Slow before the endpoint, require a stable heading tolerance, and stop between moves initially. Verify behavior physically.

Preserve the measured pose after each segment. A turn command is not proof of completion. After error-induced displacement, compute the next motion from estimated pose and check its clearance; do not blindly execute nominal lengths from the plan.

Stage C: odometry and landmarks

Update pose incrementally from encoders and gyro using the measured geometry. Begin with a declared initial pose set by physical alignment. Keep the global coordinate frame consistent even if the gyro reports relative angles.

Add seam corrections only after Test 3 passes. Only confirm the goal from black readings when the expected sensor position is near the mapped goal. Verify the required goal-arrival criterion with the instructor; the PDF does not precisely define whether the center or entire footprint must enter the goal.

Stage D: planning and execution

Store supplied obstacle centers, start and goal in config.py. Confirm workspace dimensions: the assignment PDF contains conflicting overlaid dimensions. Build a grid and your own four-neighbor BFS first; combine collinear steps into segments.

Inflate the 0.305 m square obstacles by the robot envelope plus error allowance, apply boundary clearance, and validate edges. Connect the exact start and goal safely to the grid. Test the planner on a laptop before executing a route. Reserve hardware.py, motion.py, odometry.py, planner.py, config.py and main.py for separate responsibilities; do not implement untested hardware assumptions in the planner.

Clearance and recovery decisions

Required before calling the assembled robot robust enough for the course.

Compute the available error allowance

For each route segment and turn, let c be the minimum distance from the planned robot center to any obstacle edge or workspace boundary. Using the conservative circular footprint, available position error A = c - R. If A <= 0, the route fails geometrically. In a centered 0.61 m straight corridor, c = 0.305 m.

Example only: with measured R = 0.18 m, A = 0.125 m. Choosing E = 0.05 m would leave 0.075 m reserve in that straight corridor, but only if route tests support E. None of these example values are measurements of this robot.

Account for heading, distance scale, and starting error

A constant heading error alpha over length L contributes L*sin(alpha) sideways error. At 2 m and 3 degrees, this is about 0.105 m. Add 0.02 m starting error and the illustrative 0.125 m allowance is almost entirely consumed.

A 1 percent distance-scale error contributes 0.05 m over 5 m of straight travel. Gyro feedback does not remove it. Errors propagate across turns; a square that returns near its start can still have unsafe intermediate deviations.

Measure clearance and error at intermediate checkpoints and tight turns, not just the endpoint. Include stopping overshoot, turn-center movement, and repeatable start placement in route trials. Stationary gyro drift is only one contributor.

Required adversarial route trials

Use the narrowest intended corridor, longest intended straight, most turns, and a route lasting at least as long as the planned demonstration. Test both turn directions and real seam crossings. Independently replace the robot at the start each run. Repeat at realistic battery states and room lighting.

Track all failures, including slips and false landmark detections. As a project screen, require ten consecutive full-route successes with no obstacle or boundary crossing, no manual assistance, and checkpoint error below A with chosen reserve. This is a proposed acceptance screen, not a statistical guarantee or rubric.

On failure: act on the measured source

Slip or stall: stop, mark pose uncertain, physically realign and restart. Do not resume by zeroing encoders/gyro at the unknown location. Gyro/encoder disagreement is a warning; agreement does not prove that no slip occurred.

Excess drift: correct startup and mount issues, compare speeds, and reduce avoidable turns/pauses. Persistent scale error: recalibrate. No clearance: shorten protrusions or replan. Poor seams: disable correction. Never force a landmark match to finish.

09 / Bench worksheet

Print this page / record the physical build before changing calibration.

Build record

Builder: _____ Date: _____

Battery / charge condition: _____

Floor and lighting: _____

Firmware and permitted hardware API: _____

Wheel description: _____

D_L: _____ m D_R: _____ m b: _____ m

R: _____ m s_x: _____ m s_y: _____ m

Lens gap h: _____ mm Straight speed: _____ m/s

Gyro sign: _____ Left motor sign: ____ Right motor sign: ____

Stationary gyro duration: _____ s Heading change: _____ degrees

Floor reflection range: _____ Seam range: _____ Black range: _____

Repeated trial summary

1 m forward: trials ____ Max forward error ____ Max lateral error ____

90 deg left: trials ____ Max physical angle error ____

90 deg right: trials ____ Max physical angle error ____

Square / route test: trials ____ Largest position error ____

Chosen planning error allowance E: ____ m R + E: ____ m

Seam correction enabled? ____ Evidence: _____

Before each demonstration

[ ] Tire seating, frame pins, caster, and sensor mounts checked.

[ ] Battery, floor, speed, and payload match calibration.

[ ] Course dimensions and all obstacle coordinates verified.

[ ] Planned route and turning envelope pass clearance checks.

[ ] Start center and heading aligned; alignment aids removed.

[ ] Robot still during gyro initialization; stationary readings checked.

[ ] Correct ports, stop button, timeout, and logs verified.

[ ] Goal confirmation uses location context, not black alone.

Change log

Modification: _____

Reason / observed failure: _____

Calibration and route tests repeated: _____